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B. E. (Seventh Semester) Examination, April-May 2016

(New Scheme)

(Mech. Branch)

MACHINE TOOL TECHNOLOGY

Time Allowed : Three hours

Maximum Marks : 80

Minimum Pass Marks : 28

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Note : Part (a) is compulsory from each question. Attempt any two from (b), (c) and (d). All questions carry equal marks. Explain your answer with neat sketches wherever necessary.

1. (a) What is Tool Signature? 2

(b) What are the two systems of defining the cutting

angle of a single point tool? Explain each with the help of neat sketches. 7

(c) Draw and explain Merchant Circle Diagram. State the assumptions made in development of the diagram. 7

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(d) During metal cutting test under orthogoinal cutting the following observations were made : 7

Cutting force = 1100 N

Feed force = 1020 N

Rake angle = 10°

Shear plane angle = 19°

Cutting velocity = 165 mpm

Velocity of shear = 164.5 mpm

Velocity of chip = 54.4 mpm

Show that the work input is equal to the sum of work done in shearing and work done against friction.

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2. (a) Explain the term 'Machinability Index'. 2

(b) Explain the mechanism of Tool Wear. Differentiate between face wear and flank wear. 7

(c) Either cemented carbide or ceramic may be used as tool material while machining medium carbon steel. The Taylor's tool life equation are :

(i) For Carbide $VT^{0.25} = 800$

(ii) For Oxide $VT^{1.25} = 8000$

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Calculate the break even speed at which both the tools give same tool life.

If you have to machine at speed of (i) 400 mpm (ii) 500 mpm, which one of these tools will you choose in order to have less frequent tool change. 7

(d) What are the main functions of a cutting fluid? List the essential characteristics of cutting fluid. 7

3. (a) Give three reasons why cast iron is preferred for lathe bed. 2

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(b) Analyse the forces acting on the bed under the head stock. 7

(c) Prove that the angle of inclination of V_{ee} slide way on the bed, with respect to horizontal should have a value less than 60° at prevent overturning of the saddle. 7

(d) What do you meant by Guide Ways? Discuss important point to be considered while designing guide ways. 7

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4. (a) Why spindle speeds are recommended to be arranged in Geometric Progression? 2

(b) Explain the term "Drives in Machine Tools". Discuss its classification. 7

(c) Design a six speed gear box for speed in range of 280 rpm to 900 rpm. Assume motor speed as 1440 rpm. Draw the kinematic arrangement of gears. 7

(d) Explain the following terms with reference to design of speed gear box : 7

(i) Speed loss

(ii) Kinematic diagram

5. (a) Explain the importance of feed gear box in machine tools. 2

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(b) With the help of neat sketch describe draw key drive. 7

(c) Design a feed gear box for cutting thread with pitch ranging from 1·5 mm to 9·0 mm having 12 different pitch. Use Nortan and Meander drive both. Draw the Kinematic Diagram. 7

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(d) Explain with suitable diagrams the complete test procedure to be adopted for acceptance of radial drilling machine. 7

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